Chia-Heng Liu

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Education

Cornell University — Masters, Electrical and Computer Engineering (GPA: 4.15/4.3) 2025 – 2026
Research: Mixed-Precision AI Accelerators for LLM Inference Advisor: Mohammad Alian

National Tsing Hua University — Bachelors, Materials Science and Engineering (GPA: 3.83/4.3) 2020 – 2024

Relevant Coursework: Complex Digital ASIC Design (Apple Tapeout), Multicore Architecture, Memory-Centric Computing, Computer Architecture, Data Center Architecture, Computer Networks

Honors & Awards: Phi Kappa Phi Honor Society (Top 10% of graduate students, Cornell University)

Selected Projects

ASIC Project: Stepwise-Streaming Systolic CNN Accelerator

SystemVerilog, Synopsys (VCS, Verdi, Design Compiler), ARM SRAM Compiler, PyTorch *May 2025 – Jul 2025*

- Designed a stepwise-streaming CNN accelerator with 9×8 pipelined INT8 PE mesh array in RTL.
- Implemented window buffering, weight caching, and streaming dataflow to overlap compute and memory.
- Verified accelerator pipelines using self-checking SystemVerilog testbenches with Synopsys VCS and Verdi.
- Synthesized and placed-and-routed the design in TSMC 40 nm achieving 500 MHz, 304 GOPS/s, 412.7K μm^2 area, and 5.1 TOPS/W energy efficiency.

ASIC Project: Mixed-Precision Edge LLM Accelerator (Academic Tapeout)

SystemVerilog, Synopsys (VCS, Design Compiler), PyTorch *Aug. 2025 – Present*

- Implementing SystemVerilog RTL for a mixed-precision transformer accelerator targeting LLM inference.
- Designed a 64×64 INT8 GEMM compute array with INT8×INT8 MAC datapaths and INT32 accumulation delivering 6.5 TOPS peak throughput at a 0.8 GHz target clock.
- Implemented output-stationary tiled dataflow with on-chip scratchpad buffering and double buffering for activation and weight reuse.
- Verified accelerator datapaths and nonlinear pipelines using self-checking simulation infrastructure.
- Implemented precision-scalable nonlinear units replacing MAC-based nonlinear datapaths, achieving ∼38% energy reduction and ∼5× higher area efficiency.

CPU Project: 4-Core Pipelined RISC-V CPU with NoC and Write-Back Cache

SystemVerilog, Verilator, PyMTL3 *Sep. 2025 – Nov. 2025*

- Implemented RTL for a 4-core 5-stage RISC-V processor interconnected via a router-based NoC.
- Designed hazard detection, forwarding networks, and a 2-way set-associative write-back cache with LRU.
- Verified multicore execution using PyMTL3 and Verilator, achieving 2.5× speedup on parallel workloads.

Publications

Chia-Heng Liu et al., Energy-Efficient and Accurate Stochastic Computing-based Multiply-Accumulate Architecture for Neural Network Accelerator Design (First-Author) *IEEE MCSoC 2025*

Chia-Heng Liu et al., Toward Unified and Tunable Mixed-Precision Accelerator Architectures for Low-Power Edge Inference (First-Author) *Manuscript in preparation*

Experience

Research Assistant, NYCU CERES Lab 2024–2025

- Implemented a stochastic-computing MAC architecture for neural network accelerators achieving 1049 μm^2 area and 0.645 mW power in TSMC 40 nm.
- Co-op Project with ITRI: developed a NoC-based AI accelerator simulator to evaluate low-power data movement and reuse strategies.

Technical Skills

- **Hardware:** SystemVerilog/Verilog, RISC-V, NoC, AI Accelerators, Parallel Compute Datapaths, Spice
- **EDA Tools:** Synopsys (VCS, Verdi, Design Compiler, PrimeTime), SRAM Compiler, Cadence (Innovus, Virtuoso), Mentor Calibre
- **Programming:** C++, Python, PyTorch, Linux, TCL